Problem being solved: Efficient code teaching.

The main problem is that individuals looking to start their journey in programming in a particular language (such as Python) and they wish to specialize in a particular field (such as Machine Learning).Often, these individuals must learn how to use certain libraries in that language.

The main purpose of CAI is to accelerate the learning process by packaging the information and integrating a guided AI to assist users with fundamentals required in the chosen field.

Beneficiaries:

This web application will mostly benefit newcomers to coding, as well as people looking to specialise in a particular field.

Within CAI we will also give content to users according to their skill levels and fields of interest, making it much more efficient and personalized.

Motivation for opportunity:

Currently the data scope online that one can find to learn programming for something specific is scatter across numerus websites, and this will require the user to look for good content then dive into the content.

Data collection:

GitHub, a free online platform that developers use to store code, since it has public repositories with code examples, we can utilize it.

StackOverflow with lots of people uploading codes to assist in the industry

Collection from other websites e.g. online courses and Q&A communities can be used to diversify our data collection and enhance the model.

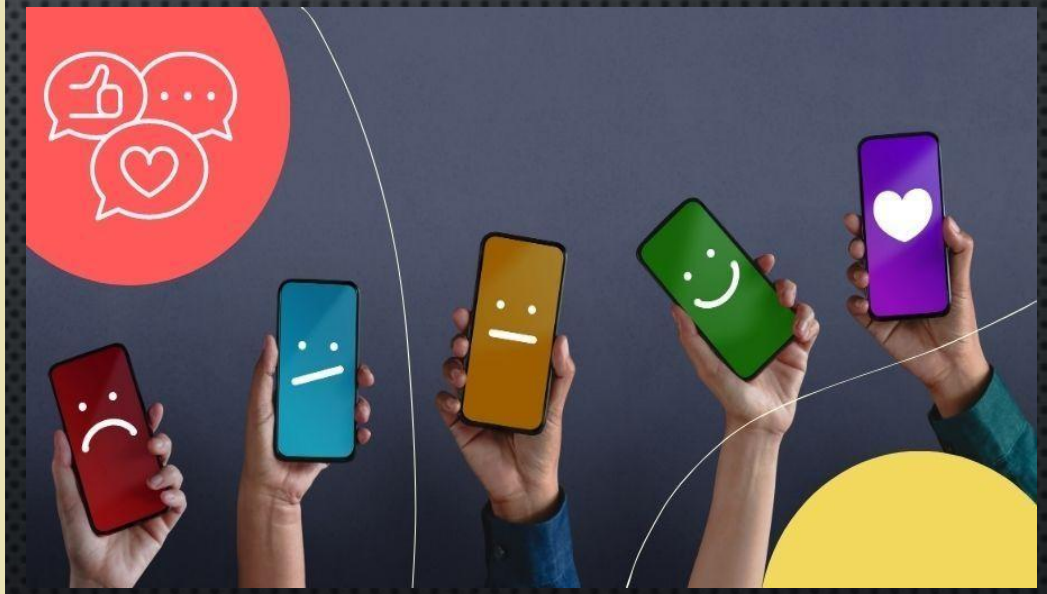
Labelling data: 3 categories

- Difficulty
- Topic
- Language

Category	Sub-category
Difficulty	Beginner
	Intermediate
	Experienced
Topic	Loops
	Classes
	Data structures
Language	Python
	Js
	Cpp

More:

To efficiently improve the models accuracy, the users will have to ability to rate suggestions, thumbs up/down and provide direct code interactions.



Data formatting:

Preprocessing:
Need to make code found in data more readable and intuitive for user.

User friendly:
Need to add comments to pieces of code if not there will need to make sure

Vast:
Need to make sure we can process code, application knows the syntax and semantics of different languages

More:

User experience:
We will also process past user experience in producing code

What kind of Machine Learning algorithm?

Supervised learning - working with, and learning from, labeled data and Reinforced via user feedback.

More Details:

- **Kind of algorithm used:**

A large language model (LLM) - a generative pre-trained transformer (GPT) to be specific. Why? Because "it's particularly well-suited for tasks like code-generation because code, like natural language, has specific syntactic and semantic rules that transformers are designed to capture".

- **Problems with over- and under-fitting**

Yes, some coding tasks might get classified as either being too difficult or too easy so this might result it under/over-fit .e.g.

More Details(cont):

too much training on specific libraries can reduce generalization (overfit), sparse labels might cause poor topic categorization (underfit)

- **Mitigation:**

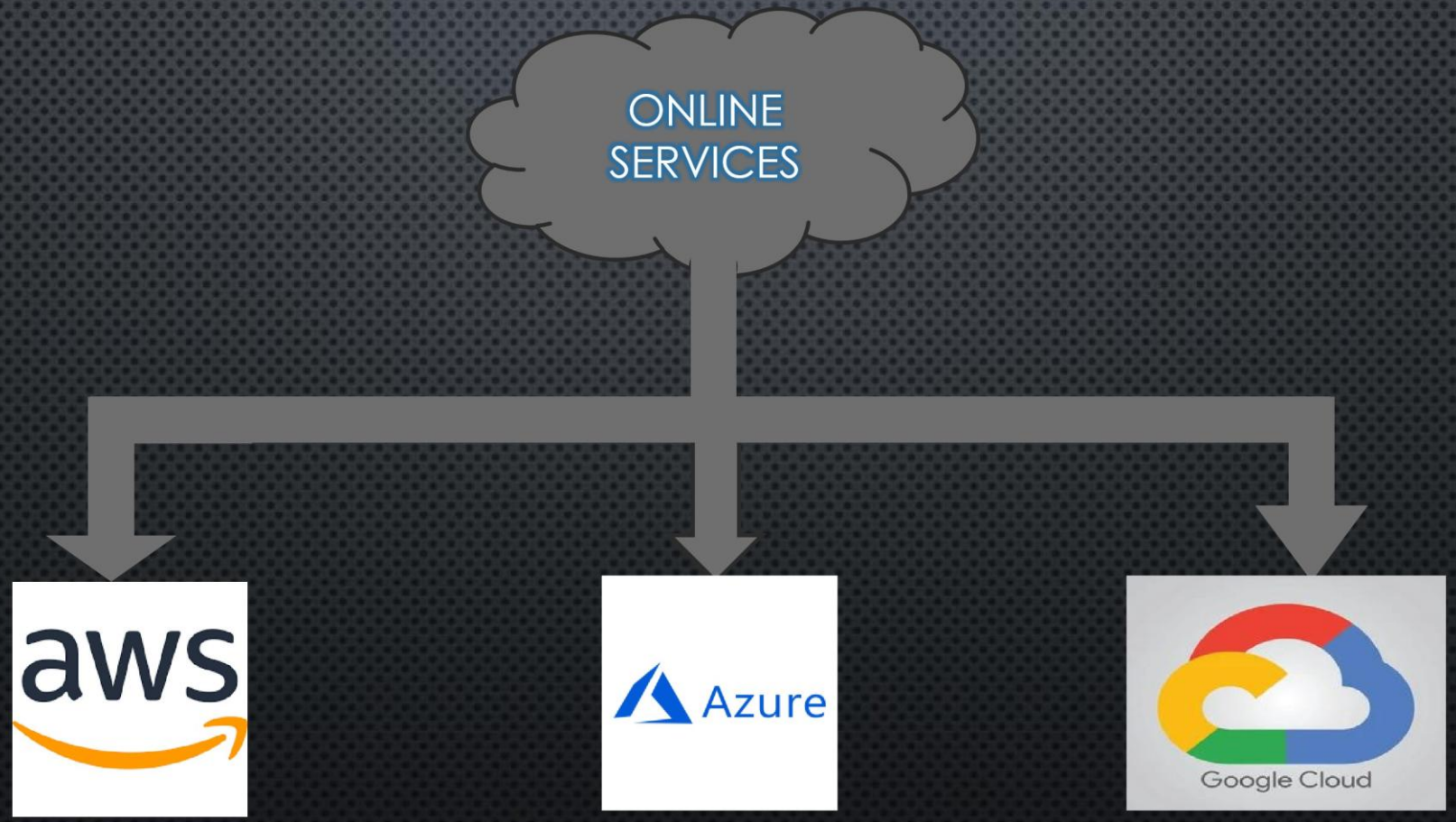
Regular cross-validation to ensure balanced learning, augmentation to get more coverage, and continual user feedback.

- **Evaluation of approach:**

We will measure success through, user feedback, track learning progression (is CAI doing good?) --> try dif models and strats

CAI deployment:

This will be a web application and it will be deployed on online services such as: (AWS, Google cloud, Azure).



Freemium	Paid tiers
Access to only one language	Access to multiple languages
Basic tutorial included	Uses ML-powered code feedback
Unlimited usage	Monthly subscription(scalable)

No ML-powered code feedback

Priority feedback

Disclaimer:

- **User Transparency:**
  All code that will be written or uploaded on our platform may be analyzed by CAI, and by so doing we will be able to give user content that they need this will also in exchange improve CAI in the real-world scenarios.

- **Purpose for Data collection:**
  The collected data will be strictly limited to only non-sensitive coding-related inputs and will be used for CAI's refinement.

  WE WILL NOT BE COLLECTING PERSONAL CODE PROJECTS OR PRIVATE CLIENT DATA UNLESS AUTHORIZED,

More:

- **Authentication Data handling:**
  Any user credentials entered(logging in <email> and <password>) will not be used for training purposes but it will only be used to authenticated user to account.

- **Use control:**
  Users will retain full control over their data. They can review/change/delete their account settings

Need some convincing??

Why invest in CAI ~ your CodeAI?

Innovative:

The use of AI in educational tools/purposes bring a change/revolution within the educational system.

Profitability:

Since the world evolves daily uses will be force to move to the premium version which will enable them to pick any programming language to their choice to learn so this will bring in huge user numbers and profits

THE END